STAT 131: Lecture Notes 6

Winter 2024

6.1 Conditional Distributions [DS 3.6]

Recall our definition of conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

We now want to extend this concept to random variables. Suppose we have a bivariate random variable $(X, Y) \sim F$, what can be said about X if Y is known?

Example 6.1 (auto insurance)**.** Suppose we have the following joint pmf:

Stolen	Brand 1	Brand 2	Brand 3	Brand 4	Brand 5	Total
No	0.129	0.298	0.161	0.280	0.108	0.976
Yes	0.010	0.010	0.001	0.002	0.001	0.024
Total	0.139	0.308	0.162	0.282	0.109	1.000

With the joint pmf, we can answer the following questions.

- What is the probability of a car being stolen? $P(X=1) = 0.024$
- What is the probability that a car is Brand 3 and is stolen? $P(X=1 \cap Y=3) = 0.001$
- Given a car is stolen, what is the probability that the stolen car is Brand 3?

$$P(Y=3 \mid X=1) = \frac{0.001}{0.01 + 0.01 + 0.001 + 0.002 + 0.001} = \frac{0.001}{0.024} \approx 0.042$$

$$= \frac{P(Y=3 \cap X=1)}{P(X=1)}$$

♢

Definition 6.1 (Conditional pmf)**.** For two discrete random variables X and Y, the conditional probability mass function (pmf) of X given $Y = y$ is

$$f_{X|Y}(x|y) = \mathbb{P}(X = x|Y = y) = \frac{f_{X,Y}(x, y)}{f_Y(y)},$$

provided $f_Y(y) > 0$.

In conditional probability, let $A = \{X = x\}$ and $B = \{Y = y\}$ and we will obtain the conditional pmf. Conditional probability calculation:

$$P(X \in E|Y = y) = \sum_{x \in E} f_{X|Y}(x|y).$$

Theorem 6.1. *The conditional pmf $f_{X|Y}(x|y)$ is a valid pmf. In other words,*

$$\sum_X f_{X|Y}(x|y) = 1.$$

Proof. Use the definition of conditional pmf.

$$\sum_x f_{X|Y}(x|y) = \sum_x \frac{f(x,y)}{f_Y(y)} = \frac{1}{f_Y(y)} \sum_x f(x,y) = \frac{1}{f_Y(y)} f_Y(y) = 1$$

□

Example 6.2. Suppose we have the following joint pmf:

	$y=1$	$y=2$	$y=3$	$y=4$	Total
$x=1$	0.1	0	0.1	0	0.2
$x=2$	0.3	0	0.1	0.2	0.6
$x=3$	0	0.2	0	0	0.2
Total	0.4	0.2	0.2	0.2	1.0

Find the conditional pmf of Y given $X = 2$.

$f_{Y|X}(y|2)$ $\quad P(Y=1|X=2) = \frac{0.3}{0.6} = \frac{1}{2}$ $\quad P(Y=2|X=2) = \frac{0}{0.6} = 0$

$P(Y=3|X=2) = \frac{0.1}{0.6} = \frac{1}{6}$ $\quad P(Y=4|X=2) = \frac{0.2}{0.6} = \frac{1}{3}$

$\sum_y f_{Y|X}(y|X=2) = \frac{1}{2} + 0 + \frac{1}{6} + \frac{1}{3} = 1$

♢

We can also deal with conditional probabilities for continuous random variables.

Definition 6.2 (Conditional pdf)**.** For two continuous random variables X and Y, the conditional probability density function pdf of X given $Y = y$ is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)},$$

provided $f_Y(y) > 0$.

Example 6.3. Visualizing a conditional pdf

♢

Using the conditional pdf, a particular conditional probability can be computed:

$$P(X \in E|Y = y) = \int_E f_{X|Y}(x|y)dx.$$

For a fixed $Y = y$, $f_{X|Y}(x|y)$ is a valid pdf of x.

Example 6.4. Suppose $f(x,y) = e^{-x}$ for $0 \le y \le x < \infty$.

- Find $f_{X|Y}(x|y)$.
- Find $\mathbb{P}(X > 9|Y = 4)$
- Find $f_{Y|X}(y|x)$.

1) $f_Y(y) = \int_y^\infty e^{-x}\,dx = -e^{-x}\Big|_y^\infty = e^{-y}$ for $0 \le y$

$\Rightarrow f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{e^{-x}}{e^{-y}}$ for $y \le x < \infty$

2) $P(X > 9 \mid Y=4) = \int_9^\infty f_{X|Y}(x|y=4)\,dx = \int_9^\infty \frac{e^{-x}}{e^{-4}}\,dx$

$= \frac{-e^{-x}}{e^{-4}}\Big|_9^\infty = e^{-9}/e^{-4} = e^{-5}$

3) $f_X(x) = \int_0^x e^{-x}\,dy = ye^{-x}\Big|_0^x = xe^{-x}$

$\Rightarrow f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{e^{-x}}{xe^{-x}} = \frac{1}{x}$ $0 \le y \le x$

♢

For conditional distribution functions, similar conclusions to those of conditional probability also hold.

Theorem 6.2 (Multiplication rule)**.**

$$f_{(X,Y)}(x,y) = f_{X|Y}(x|y)f_Y(y) = f_{Y|X}(y|x)f_X(x).$$

Example 6.5. Suppose $X \sim Unif[0,1]$, $Y|X = x \sim Unif[x,1]$.

- Find the joint pdf $f(x,y)$.
- Find the marginal distribution of Y.

1) $f(x,y) = f_{Y|X}(y|x) \cdot f_X(x) = \frac{1}{1-x} \cdot 1 = \frac{1}{1-x}$ for $0 < x < y < 1$

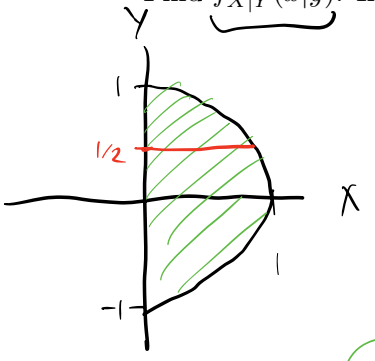
2) $f_Y(y) = \int_0^y f(x,y)\,dx = \int_0^y (1-x)^{-1}\,dx = -\log(1-x)\Big|_0^y = -\log(1-y)$ for $0 < y < 1$

♢

Example 6.6. Suppose (X,Y) is uniformly distributed on the semicircle

$$\{(x,y) : x^2 + y^2 \le 1, x \ge 0\}. \quad x \le \sqrt{1-y^2}$$

Find $f_{X|Y}(x|y)$. In particular, what is $f_{X|Y}(x|0)$?



$f(x,y) = c$ $\quad f_Y(y) = \int_0^{\sqrt{1-y^2}} c\,dx = cx\Big|_0^{\sqrt{1-y^2}} = c\sqrt{1-y^2}$

$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{c}{c\sqrt{1-y^2}} = \frac{1}{\sqrt{1-y^2}}$ for $0 \le x \le \sqrt{1-y^2}$

$f_{X|Y}(x|y=0) = \frac{1}{\sqrt{1-0^2}} = 1$ for $0 \le x \le 1$

♢

Theorem 6.3 (Law of total probability)**.** *Suppose Y is continuous,*

$$f_X(x) = \int f_{(X,Y)}(x,y)\,dy = \int f_{X|Y}(x|y)f_Y(y)\,dy.$$

Suppose Y is discrete,

$$f_X(x) = \sum_y f_{(X,Y)}(x,y) = \sum_y f_{X|Y}(x|y)f_Y(y).$$

Theorem 6.4 (Bayes theorem)**.** *Given two random variables X and Y, we can find the conditional pdf/pmf of X given $Y = y$ as*

$$f_{X|Y}(x,y) = \frac{f_{Y|X}(y|x)f_X(x)}{f_Y(y)}$$

where $f_Y(y) > 0$.

Example 6.7. Suppose we have the following conditional pmf

$$f_{Y|X}(y|x) = \frac{x^2+y^2}{2x^2+1}, \text{ for } y = 0,1$$

and the following two marginal pmfs

$$f_Y(y) = \frac{14+3y^2}{31}, \text{ for } y = 0,1$$

$$f_X(x) = \frac{2x^2+1}{31}, \text{ for } x = 1,2,3.$$

Find the conditional distribution of X given Y and show that this is a valid distribution.

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)\cdot f_X(x)}{f_Y(y)} = \frac{\frac{x^2+y^2}{2x^2+1}\cdot\frac{2x^2+1}{31}}{\frac{14+3y^2}{31}} = \frac{x^2+y^2}{2x^2+1}\cdot\frac{2x^2+1}{31}\cdot\frac{31}{14+3y^2} = \frac{x^2+y^2}{14+3y^2}$$

for $x = 1,2,3$

♢

Theorem 6.5. *$X \perp Y$ iff $f_{X|Y}(x|y) = f_X(x)$ with $f_Y(y) > 0$.*

Proof. Use definition of conditional distribution.

($\Rightarrow$) Let $X \perp Y$. Then $f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{f_X(x)\cdot f_Y(y)}{f_Y(y)} = f_X(x)$ ✓

($\Leftarrow$) Let $f_{X|Y}(x|y) = f_X(x) \Rightarrow \frac{f(x,y)}{f_Y(y)} = f_X(x)$

$\Rightarrow f(x,y) = f_X(x)\cdot f_Y(y) \Rightarrow X \perp Y$ ✓

□

6.2 Transformations of Random Variables [DS 3.8]

In many cases, we know the distribution of some random variable X, but we really care about some transformation of function of X. For example if X is the rate that customers are lining up in a queue, then $1/X$ corresponds to the average waiting time. If we know the distribution of X, we should also be able to find the distribution of $1/X$, which is the problem we will study now.

Example 6.8 (Distance from the middle)**.** Let X be a random variable that follows a uniform distribution on the integers $1, 2, \ldots, 9$. Suppose that we are interested in how for X is from the middle of the distribution (i.e., 5). We could define $Y = |X - 5|$, and then consider probabilities involving Y. What is $\mathbb{P}(Y = 1)$?

$P(Y=1) = P(X = \{4, 6\}) = 1/9 + 1/9 = 2/9$

♢

Theorem 6.6 (Transformation of a discrete random variable)**.** *Let X have a discrete distribution with pmf $f_X(x)$ and let $Y = r(X)$ for some function of $r(\cdot)$ defined on the support of X. For each possible value y of Y, the pmf of Y is*

$$f_Y(y) = \mathbb{P}(Y = y) = \mathbb{P}(r(X) = y) = \sum_{x:r(x)=y} f_X(x).$$

Example 6.9 (Distance from the middle)**.** Let X be a random variable that follows a uniform distribution on the integers $1, 2, \ldots, 9$ and $Y = |X - 5|$. What is the pmf of Y?

Possible values of Y: $0, 1, 2, 3, 4$

$P(Y=0) = P(X=5) = 1/9$ $\quad P(Y=1) = 2/9$

$P(Y=2) = P(X \in \{3, 7\}) = 2/9$ $\quad P(Y=3) = P(X \in \{2, 8\}) = 2/9$

$P(Y=4) = P(X \in \{1, 9\}) = 2/9$ $\quad \sum_y f_Y(y) = 1/9 + 2/9 + 2/9 + 2/9 + 2/9 = 1$

♢

We can also consider transformations in the continuous case, however for continuous random variables, it is often more convenient to work with the cdf.

Theorem 6.7 (Transformation of a continuous random variable)**.** *Let X have a continuous distribution with cdf $F_X(x)$ and let $Y = r(X)$ for some function of $r(\cdot)$ defined on the support of X. Then, the cdf of Y is*

$$F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}(r(X) \le y) = \int_{x:r(x)\le y} f_X(x)dx.$$

Example 6.10 (Average waiting time)**.** Let Z be the rate at which customers are served in a queue and assume that $Z \sim \text{Exp}(\lambda)$. The average waiting time is $Y = 1/Z$. Find the cdf and pdf of Y.

$F_Z(z) = 1 - e^{-\lambda z}$ we want $F_Y(y) = P(Y \le y)$ Possible values of y? $y > 0$

$$P(Y \le y) = P\left(\frac{1}{Z} \le y\right) = P(Z \ge 1/y)$$

$$= 1 - P(Z \le 1/y) = 1 - F_Z(1/y) = 1 - (1 - e^{-\lambda/y}) = e^{-\lambda/y} \quad \text{for } y > 0$$

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} e^{-\lambda/y} = e^{-\lambda/y} \cdot \lambda \cdot y^{-2} = \frac{\lambda e^{-\lambda/y}}{y^2} \quad \text{for } y > 0$$

◊

Example 6.11. Suppose that X follows a uniform distribution on the interval $[-1, 1]$. Find the pdf of the random variable $Y = X^2$.

$f_X(x) = 1/2$ for $-1 \le x \le 1$ $\Rightarrow \int_{-1}^{x} \frac{1}{2}\, dt = \frac{x}{2}\Big|_{-1}^{x} = \frac{x+1}{2}$ Possible values of y? $0 \le y \le 1$

$$F_Y(y) = P(Y \le y) = P(X^2 \le y) = P(-\sqrt{y} \le x \le \sqrt{y}) = P(x \le \sqrt{y}) - P(x \le -\sqrt{y}) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

$$= \frac{y^{1/2}+1}{2} - \left(\frac{-y^{1/2}+1}{2}\right) = \frac{2y^{1/2}}{2} = y^{1/2} \quad \text{for } 0 \le y \le 1$$

$$\Rightarrow f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} y^{1/2} = \frac{1}{2} y^{-1/2} \quad \text{for } 0 \le y \le 1$$

◊

Theorem 6.8 (Linear transformation)**.** *Suppose X is a continuous random variable with pdf $f_X(x)$ and $Y = aX + b$ for $a \neq 0$. Then the pdf of Y is*

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{Y - b}{a}\right)$$

$y = \frac{1}{5}x + 0 \quad a = \frac{1}{5} \quad b = 0$

Example 6.12. Suppose $X \sim \text{Exp}(1)$. Find the pdf of $Y = \frac{X}{5}$.

$f_X(x) = e^{-x} \qquad f_Y(y) = \frac{1}{1/5} f_X\left(\frac{y-0}{1/5}\right) = 5e^{-5y}$

$\Rightarrow Y \sim \text{Exp}(5)$ ♢

Example 6.13. Suppose $X \sim \text{Exp}(1)$. Then $f_X(x) = e^{-x}$ and $F_X(x) = 1 - e^{-x}$ for $x > 0$. Find the pdf of $Y = F_X(x)$.

$F_Y(y) = P(Y \le y) = P(F_X(x) \le y) = P\left(F_X^{-1}F_X(x) \le F_X^{-1}(y)\right)$

$= P(X \le F_X^{-1}(y)) = F_X\left(F_X^{-1}(y)\right) = y$

$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} y = 1$ ♢

Theorem 6.9 (Probability integral transformation)**.** *Suppose X is a continuous random variable with cdf $F_X(x)$ and $Y = F_X(X)$. Then $Y \sim Unif[0,1]$.*

This theorem implies that we can use an inverse CDF method in random number simulation. Suppose $Y \sim Unif[0,1]$, and $F(\cdot)$ is a continuous CDF with quantile function $F^{-1}(\cdot)$. Then $F^{-1}(Y) \sim F$. To be more specific, the following is an algorithm to simulate random numbers from a distribution with cdf $F(\cdot)$:

(1) first generate random numbers U_i's from $Unif[0,1]$;

(2) then, apply inverse-CDF transform $F^{-1}(U_i)$.

The resulting random numbers $\{F^{-1}(U_i)\}_{i=1}^n$ follow the distribution F.